

Protocol 1 II.IDEq (Identity-Equality Proof)

Parameter: Security parameter λ . Public parameters $\text{pp} \leftarrow \text{Setup}(1^\lambda)$: a group $\mathbb{G}$ of prime order p with generators $g_{\text{id}}, g_{\text{val}}, g_{\text{ran}} \in \mathbb{G}$ produced by `HASHToGroup` from fixed domain strings (Algorithm 0), so their mutual discrete-log relations are unknown. [P3]

Input: (ID, value) commitments $c_b = g_{\text{id}}^{\text{id}_b} \cdot g_{\text{val}}^{v_b} \cdot g_{\text{ran}}^{r_b}$ for $b = 0, 1$. The prover holds the openings (id_b, v_b, r_b) ; the verifier holds c_0, c_1 and the context string ctx . [P2]

Output: Completeness with probability 1; knowledge soundness with knowledge error $1/p$ under the DLR assumption on $(g_{\text{id}}, g_{\text{val}}, g_{\text{ran}})$; honest-verifier zero-knowledge (interactive) and zero-knowledge in the random-oracle model (non-interactive). The conclusion $\text{id}_0 = \text{id}_1$ follows only under DLR (Theorem 1). [P1]

Interactive Proof

Prover

1: Randomly sample $\alpha_{\text{val}}, \alpha_{\text{ran}} \in \mathbb{Z}_p$
2: Compute $t = g_{\text{val}}^{\alpha_{\text{val}}} \cdot g_{\text{ran}}^{\alpha_{\text{ran}}}$
 t
3: Randomly sample $\beta \in \mathbb{Z}_p$
 β
4: Compute $z_{\text{val}} = \beta(v_0 - v_1) + \alpha_{\text{val}}$
 $z_{\text{ran}} = \beta(r_0 - r_1) + \alpha_{\text{ran}}$
 $z_{\text{val}}, z_{\text{ran}}$

Verifier

5: **if** $g_{\text{val}}^{z_{\text{val}}} \cdot g_{\text{ran}}^{z_{\text{ran}}} = t \cdot (c_0/c_1)^\beta$ **then**
6: Return 1 (accept)
7: **else**
8: Return 0 (reject)

Non-Interactive Proof (Fiat-Shamir)

Prover

1: Randomly sample $\alpha_{\text{val}}, \alpha_{\text{ran}} \in \mathbb{Z}_p$
2: Compute $t = g_{\text{val}}^{\alpha_{\text{val}}} \cdot g_{\text{ran}}^{\alpha_{\text{ran}}}$
3: Compute $\beta = H(\text{ds}_{\text{IDeq}}, \text{pp}, c_0, c_1, \text{ctx}, t)$
4: Compute $z_{\text{val}} = \beta(v_0 - v_1) + \alpha_{\text{val}}$
 $z_{\text{ran}} = \beta(r_0 - r_1) + \alpha_{\text{ran}}$
5: Set $\pi = (t, z_{\text{val}}, z_{\text{ran}})$
 π

Verifier

6: Compute $\beta = H(\text{ds}_{\text{IDeq}}, \text{pp}, c_0, c_1, \text{ctx}, t)$
7: **if** $g_{\text{val}}^{z_{\text{val}}} \cdot g_{\text{ran}}^{z_{\text{ran}}} = t \cdot (c_0/c_1)^\beta$ **then**
8: Return 1 (accept)
9: **else**
10: Return 0 (reject)

The proof $\pi = (t, z_{\text{val}}, z_{\text{ran}})$ is one group element and two scalars — 96 bytes on ristretto255. No β is exchanged; the verifier recomputes it from the same public inputs.

Protocol 2 II.VVer (Value-Verification Proof)

Parameter: Security parameter λ ; group parameters as in Protocol 1 (generators from `HASHToGroup`, Algorithm 0). [P3]

Input: The prover holds id, v, r and $c = g_{\text{id}}^{\text{id}} \cdot g_{\text{val}}^v \cdot g_{\text{ran}}^r$; the verifier holds the public value v' , the commitment c , and the context string ctx . [P2]

Output: Completeness with probability 1; knowledge soundness with knowledge error $1/p$ under DLR; honest-verifier zero-knowledge (interactive) and zero-knowledge in the random-oracle model (non-interactive). The conclusion $v = v'$ follows only under DLR (Theorem 2). [P1]

Interactive Proof

Prover

1: Randomly sample $\alpha_{\text{id}}, \alpha_{\text{ran}} \in \mathbb{Z}_p$
2: Compute $t = g_{\text{id}}^{\alpha_{\text{id}}} \cdot g_{\text{ran}}^{\alpha_{\text{ran}}}$
 t
3: Randomly sample $\beta \in \mathbb{Z}_p$
 β
4: Compute $z_{\text{id}} = \beta \cdot \text{id} + \alpha_{\text{id}}$
 $z_{\text{ran}} = \beta \cdot r + \alpha_{\text{ran}}$
 $z_{\text{id}}, z_{\text{ran}}$

Verifier

5: **if** $g_{\text{id}}^{z_{\text{id}}} \cdot g_{\text{ran}}^{z_{\text{ran}}} = t \cdot (c \cdot g_{\text{val}}^{-v'})^\beta$ **then**
6: Return 1 (accept)
7: **else**
8: Return 0 (reject)

Non-Interactive Proof (Fiat-Shamir)

Prover

1: Randomly sample $\alpha_{\text{id}}, \alpha_{\text{ran}} \in \mathbb{Z}_p$
2: Compute $t = g_{\text{id}}^{\alpha_{\text{id}}} \cdot g_{\text{ran}}^{\alpha_{\text{ran}}}$
3: Compute $\beta = H(\text{ds}_{\text{VVer}}, \text{pp}, c, v', \text{ctx}, t)$
4: Compute $z_{\text{id}} = \beta \cdot \text{id} + \alpha_{\text{id}}$
 $z_{\text{ran}} = \beta \cdot r + \alpha_{\text{ran}}$
5: Set $\pi = (t, z_{\text{id}}, z_{\text{ran}})$
 π

Verifier

6: Compute $\beta = H(\text{ds}_{\text{VVer}}, \text{pp}, c, v', \text{ctx}, t)$
7: **if** $g_{\text{id}}^{z_{\text{id}}} \cdot g_{\text{ran}}^{z_{\text{ran}}} = t \cdot (c \cdot g_{\text{val}}^{-v'})^\beta$ **then**
8: Return 1 (accept)
9: **else**
10: Return 0 (reject)

The public value v' is bound into β : the verified element $c \cdot g_{\text{val}}^{-v'}$ moves with v' , so omitting it would let a proof for one value be claimed for another. Proof size 96 bytes.